Wheat Cultivation Guide 2024

Comprehensive Advisory for Rabi Season

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1. Introduction

Wheat is the second most important cereal crop of India after rice, contributing substantially to the national food security. India is the second largest producer of wheat in the world. Wheat is primarily a rabi season crop sown during October-November and harvested during March-April. The Indo-Gangetic plains of North India comprising Punjab, Haryana, and Uttar Pradesh account for about 65% of the total wheat production in the country.

1.1 Recommended Varieties for North India

The following improved wheat varieties are recommended for cultivation in the North-Western Plains Zone covering Punjab, Haryana, and western Uttar Pradesh:

- **HD-2967**: High yielding variety suitable for irrigated timely sown conditions. Average yield potential 50-55 q/ha. Resistant to yellow rust and leaf rust.
- **PBW-343**: Widely adapted variety for irrigated conditions. Semi-dwarf with good tillering capacity. Average yield 45-48 q/ha.
- **DBW-17**: Recommended for late sown irrigated conditions. Heat tolerant variety with yield potential of 40-45 q/ha.
- **WH-1105**: Suitable for restricted irrigation conditions in Haryana and Punjab. Good grain quality with bold grains.

2. Soil and Land Preparation

Wheat grows best in well-drained loamy to clay loam soils with good organic matter content. The ideal soil pH range for wheat cultivation is 6.0 to 7.5. Saline and alkaline soils with pH above 8.5 should be avoided or reclaimed before sowing.

Land preparation involves 2-3 ploughings with a cultivator followed by planking to achieve a fine tilth. Laser leveling of fields is recommended for uniform irrigation and better crop stand. Incorporate FYM or compost at 10-15 tonnes per hectare during the last ploughing for improved soil fertility and structure.

Conservation tillage practices like zero tillage can save 1500-2000 rupees per hectare in land preparation costs while maintaining or improving yield levels. Zero till wheat also helps in timely sowing after rice harvest in the rice-wheat cropping system.

3. Seed and Sowing

3.1 Seed Rate and Treatment

The recommended seed rate for wheat is 100 kg per hectare for timely sown conditions using a seed drill. For late sown conditions, increase the seed rate to 125 kg/ha to compensate for reduced tillering. Use only certified seed with germination above 85%.

Seed treatment is essential for protection against seed-borne diseases. Treat seed with Carboxin or Thiram at the rate of 2.5 g per kg of seed. For loose smut management, treat with Carbendazim at 2.5 g per kg seed or Tebuconazole 2DS at 1.5 g per kg seed. Seed treatment with Imidacloprid 600FS at 1.5 ml per kg seed protects against termites.

3.2 Sowing Time and Method

The optimum sowing time for wheat in North India is October 25 to November 25 for timely sown varieties. Late sown varieties can be planted up to December 25 with appropriate adjustments in seed rate and variety selection. Sowing should be done in rows spaced 20 to 22.5 cm apart using a seed drill at a depth of 4-5 cm for proper germination and uniform crop stand.

4. Fertilizer Management

The recommended fertilizer dose for irrigated wheat is 120:60:40 kg N:P:K per hectare. Apply half of the nitrogen (60 kg N/ha) along with the full dose of phosphorus (60 kg P2O5/ha) and full potassium (40 kg K2O/ha) as basal dose at the time of sowing.

The remaining nitrogen should be applied in two equal splits: first top dressing of 30 kg N/ha at the time of first irrigation (21 DAS, crown root initiation stage), and second top dressing of 30 kg N/ha at the time of second irrigation (tillering stage). Use urea as the nitrogen source for top dressing.

For zinc deficient soils, apply zinc sulphate at 25 kg per hectare at the time of sowing. Apply FYM or farm yard manure at 10 to 15 tonnes per hectare 3-4 weeks before sowing. Green manuring with dhaincha or sunhemp before wheat sowing can supplement 20-30 kg N/ha and improve soil organic carbon.

5. Irrigation Management

Wheat requires 5 to 6 irrigations for optimum production. The critical stages for irrigation in wheat are as follows:

- **Crown Root Initiation (CRI)**: First irrigation at 21 days after sowing (DAS). This is the most critical irrigation and should not be delayed.
- **Tillering Stage**: Second irrigation at 40-45 DAS for promoting tiller development.
- **Jointing Stage**: Third irrigation at 60-65 DAS when nodes begin to form.
- **Flowering Stage**: Fourth irrigation at 80-85 DAS during anthesis.
- **Milk Stage**: Fifth irrigation at 100-105 DAS when grains are in milk stage.
- **Dough Stage**: Sixth irrigation at 115-120 DAS during grain filling.

If water availability is limited, prioritize irrigations at CRI, flowering, and milk stages. Sprinkler irrigation can save 20-30% water compared to flood irrigation with comparable yields.

6. Pest and Disease Management

6.1 Major Diseases

• **Yellow Rust (Stripe Rust)**: Most destructive disease of wheat in North India. Appears as yellow-orange pustules in stripes on leaves. Spray Propiconazole 25EC at the rate of 1 ml per litre of water as soon as symptoms appear. Repeat after 15 days if disease persists.

• **Karnal Bunt**: Affects individual grains in the ear head, producing black powdery mass with fishy smell. Spray Propiconazole 25EC at boot stage for preventive control. Use disease-free certified seed.

• **Loose Smut**: Entire ear head is replaced by black powdery spore mass. Manage through seed treatment with Carbendazim or Carboxin at 2.5 g/kg seed.

6.2 Major Insect Pests

• **Termites**: Damage roots and lower stem causing drying of plants in patches. Apply Chlorpyrifos 20EC at 4 ml per litre of water as soil drench near the affected plants. Seed treatment with Imidacloprid also provides early protection.

• **Aphids**: Suck sap from leaves and ear heads during grain filling stage. Spray Dimethoate 30EC at 1 ml per litre of water or Imidacloprid 17.8SL at 0.5 ml per litre when aphid population exceeds economic threshold level.

7. Expected Yield

Under recommended management practices and timely sown irrigated conditions, the expected yield of improved wheat varieties is 45 to 50 quintals per hectare. Late sown wheat yields 35-40 q/ha. Under rainfed conditions in North India, wheat yield ranges from 20-25 q/ha depending on rainfall distribution. Adoption of all recommended practices including balanced fertilization, timely irrigation, and integrated pest management can help achieve the maximum potential yield.